Prolog - Conjunctions & Disjunctions

1. Conjunction (Logical AND)

- Conjunction also known as Logical AND can be implemented using the comma (,) operator. So two predicates separated by comma are joined with AND statement.

- Suppose we have a predicate, **parent(jhon, bob)**, which means “Jhon is the parent of Bob”, and another predicate, **male(jhon)**, which means “Jhon is male”. So we can make another predicate that **father(jhon,bob)**, which means “Jhon is father of Bob”.

- We can define predicate **father**, when he is parent **AND** he is male.

2. Disjunction (Logical OR)

- Disjunction (Logical OR) can be implemented using the semi-colon (;) operator. So two predicates separated by semi-colon are joined with OR statement.
- Suppose we have a predicate, **father(jhon, bob)**. This tells that “Jhon is father of Bob”, and another predicate, **mother(lili,bob)**, this tells that “lili is mother of bob”.
- If we create another predicate as **child()**, this will be true when **father(jhon, bob)** is true **OR mother(lili,bob)** is true.

Try the following

```
parent(jhon,bob).
parent(lili,bob).

male(jhon).
female(lili).

% Conjunction Logic
father(X,Y) :- parent(X,Y),male(X).
mother(X,Y) :- parent(X,Y),female(X).

% Disjunction Logic
child_of(X,Y) :- father(X,Y);mother(X,Y).
```

Query 1: father(jhon,bob).
Output
yes

Query 2: child_of(jhon,bob).
Output
true ?

Query 3: father(jhon,bob).
Output
yes

Query 4: child_of(lili,bob).
Output
yes

Review Exercises

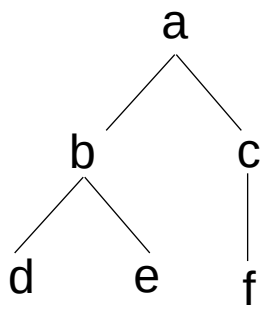
Q1. Try the following Prolog program and try the sample query below.

Assume given a set of facts of the form father(name1,name2) (name1 is the father of name2).

a) Define a predicate brother(X,Y) which holds iff X and Y are brothers.
b) Define a predicate cousin(X,Y) which holds iff X and Y are cousins.
c) Define a predicate grandson(X,Y) which holds iff X is a grandson of Y.
d) Define a predicate descendent(X,Y) which holds iff X is a descendent of Y.
e) Consider the following genealogical tree:

```
father(a,b).  % 1
father(a,c).  % 2
father(b,d).  % 3
father(b,e).  % 4
father(c,f).  % 5
```

whose graphical representation is:



Say which answers, and in which order, are generated by your definitions for the queries

```
?- brother(X,Y).
?- cousin(X,Y).
?- grandson(X,Y).
?- descendent(X,Y).
```